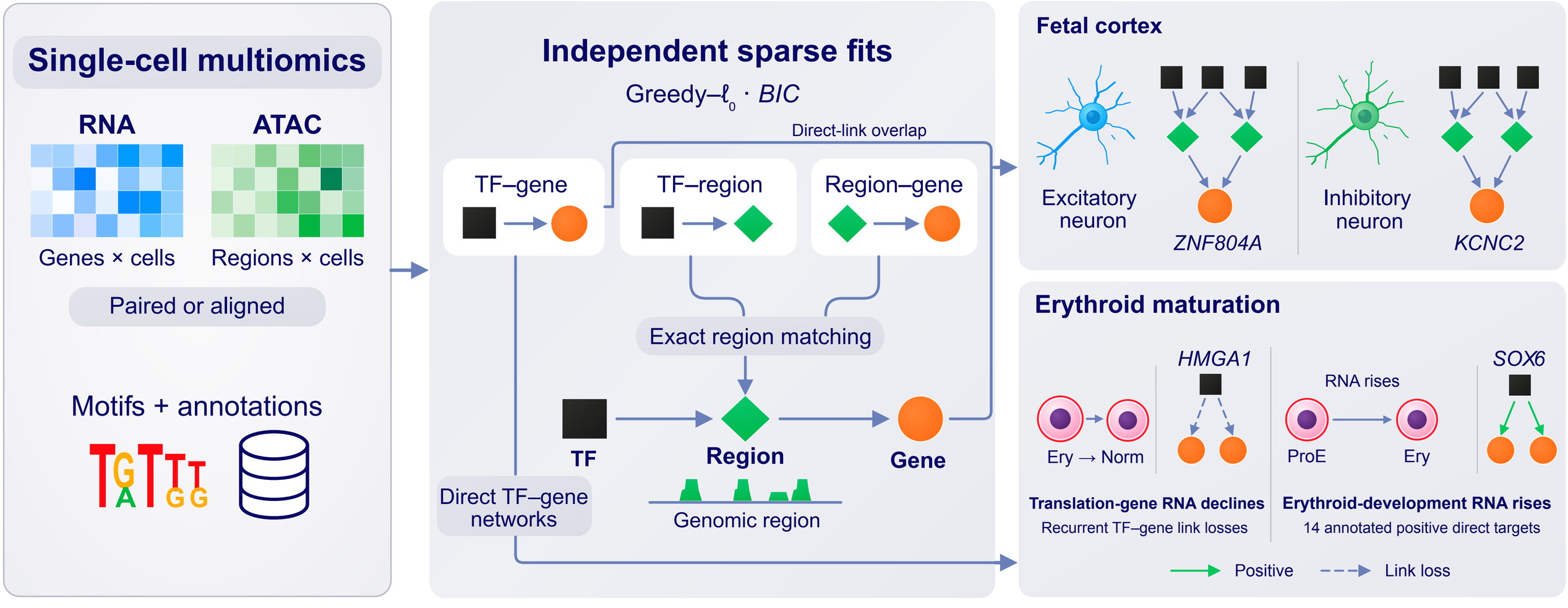

Single-cell multiomics
RNA
ATAC
Genes × cells
Regions × cells
Paired or aligned
Motifs + annotations
Independent sparse fits
Greedy–ℓ_0 · BIC
Direct-link overlap
TF–gene
TF–region
Region–gene
Exact region matching
TF
Region
Gene
Genomic region
Direct TF–gene networks
Fetal cortex
Excitatory neuron
ZNF804A
Inhibitory neuron
KCNC2
Erythroid maturation
HMGA1
SOX6
RNA rises
Ery → Norm
ProE
Ery
Translation-gene RNA declines
Recurrent TF–gene link losses
Erythroid-development RNA rises
14 annotated positive direct targets
Positive
Link loss